



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination 2018
Higher 1

CANDIDATE
NAME

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CIVICS
GROUP

1	7				
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INDEX
NUMBER

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H1 BIOLOGY

8876/02

Paper 2

12 September 2018

2 hours

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **ONE** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/ 9
2	/ 10
3	/ 11
4	/ 15
Section B	
5 or 6	/ 15
Total	/ 60

Section A
Answer **all** questions in this section

QUESTION 1

(a) Distinguish the processes of facilitated diffusion and active transport. [3]

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(b) A group of students investigated the uptake of chloride ions in barley plants. They divided the plants into two groups and placed their roots in solutions containing radioactive chloride ions.

- Group **A** plants had a substance that inhibited respiration added to the solution.
- Group **B** did not have the substance added to the solution.

The students calculated the total amount of chloride ions absorbed by the plants every 15 minutes. Their results are shown in Fig. 1.1.

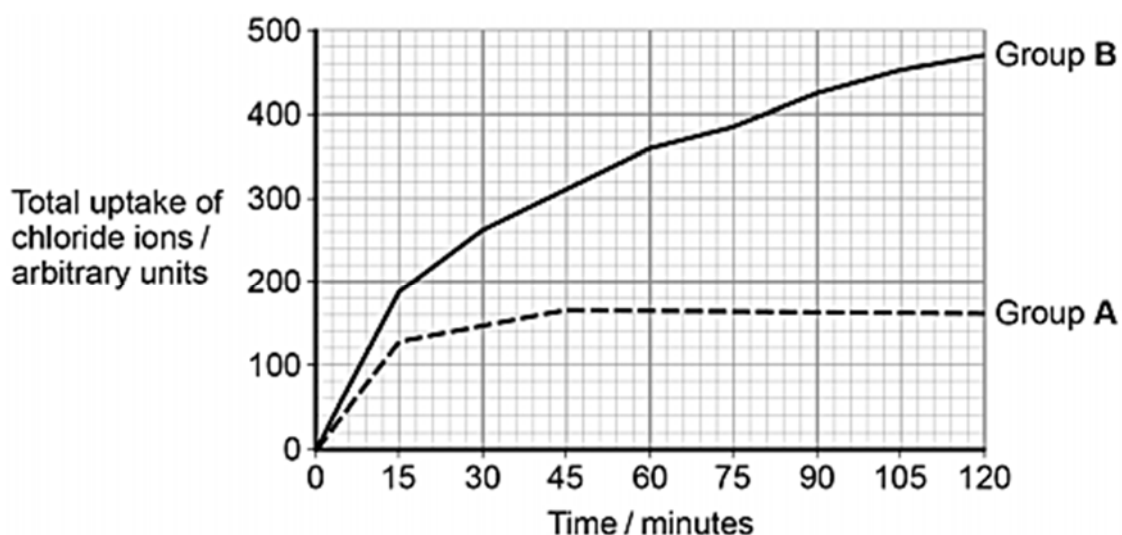


Fig. 1.1

- (i) Calculate the ratio of the **rate** of uptake of chloride ions in the first hour to the **rate** of uptake of chloride ions in the second hour for group **B** plants. [2]

- (ii) Explain the results shown in Fig 1.1. [4]

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[Total: 9]

QUESTION 2

Bone marrow contains many stem cells. Some of these stem cells are responsible for the replacement of white blood cells.

Fig. 2.1 shows the production of a white blood cell from one of these stem cells.

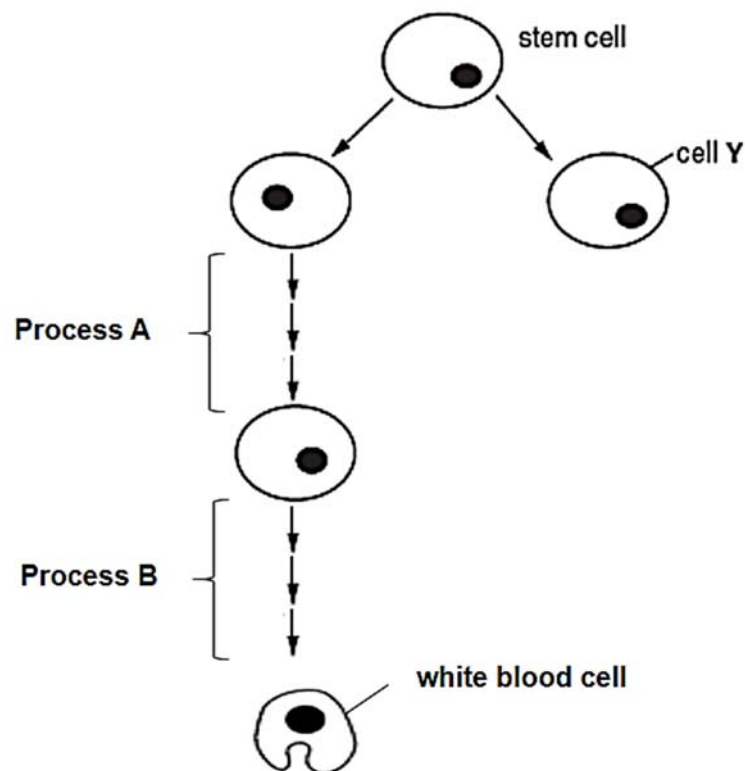


Fig. 2.1

(a) (i) State the potency of the stem cells found in the bone marrow. [1]

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(ii) Name the processes A and B. [2]

A

B

(iii) Suggest what may happen to cell Y. [1]

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The human induced pluripotent stem cells (HiPS cells) can be derived from fully differentiated adult somatic cells such as white blood cells. This has been hailed as an effective replacement for human embryonic stem cells for its usefulness in regenerative medicine.

Recent research shows that abnormal reprogramming can occur during the induction of HiPS cells and tumours could be generated.

(b) (i) Compare the embryonic stem cell and HiPS cell. **[3]**

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(ii) Comment on the ethical issues involved in the use of HiPS cells. **[3]**

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[Total: 10]

QUESTION 3

In mice, fur colour is controlled by a gene with multiple alleles. These alleles are listed below in no particular order.

black and tan = C^{bt}
agouti = C^a

yellow = C^y
black = C^b

(a) Explain the term *multiple alleles*.

[2]

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(b) Crosses between heterozygous parents with the genotype C^yC^b always produce a ratio of two yellow mice to one black mouse. Explain the observation using a genetic diagram.

[3]

(c) Suggest explanations for the results of the following crosses between mice.

- (i)** Mice with agouti fur crossed with mice with black fur may produce all agouti offspring **or** some agouti and some black offspring. [2]

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- (ii)** Mice with yellow fur crossed with mice with black fur will produce one of the following outcomes:

- some yellow offspring and some agouti offspring
- some yellow offspring and some black and tan offspring
- some yellow offspring and some black offspring. [2]

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(d) A test cross is used to determine the genotype of an organism.

Describe how you would carry out a test cross to determine the genotype of a black and tan mouse. [2]

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[Total: 11]

QUESTION 4

Malaria is a mosquito-borne infectious disease caused by the malarial parasites, *Plasmodium falciparum*, which is transmitted among humans by the female mosquito vectors *Anopheles*.

(a) Outline the general life cycle of a mosquito. [3]

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Malaria was common in Italy, a European country situated in the Northern Hemisphere. Widespread land drainage together with the use of the insecticide DDT and the drug chloroquine eradicated both the mosquito vectors and the malarial parasites in the 1950s. Due to the success of these measures, they were later discontinued as they were no longer necessary.

Articles in the scientific literature more recently show that malarial mosquitoes are returning to Italy and increasing their numbers and their northerly range, with some cases of malaria being reported. In general, winters are milder and summers are hotter in the south of the country, with temperatures decreasing to the north, especially in the winters.

(b) Discuss whether the return of malaria to Italy can be attributed to climate change. [4]

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- (c) Chloroquine-resistant (CQR) malarial parasites, *P. falciparum*, were first reported in 1950s and are now widespread. The resistance is caused by mutations of a gene known as *pfcr*.

(i) Explain why CQR *P. falciparum* are now widespread. [3]

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Analysis of *pfcr* in CQR *P. falciparum* from different parts of the world shows differences in one section of the gene only. The amino acid sequences coded for by this section are shown in Table 4.1, together with the amino acid sequence coded for by the allele found in parasites that are still susceptible to chloroquine (non-CQR). The amino acid sequence coded for by the rest of the gene is the same in all cases.

Table 4.1

allele of <i>pfcr</i> gene	amino acid sequence coded for by allele of <i>pfcr</i> gene
non-CQR	—Cys—Met—Asn—Lys—His—Ala—Glu—Asn—Ile —Met—
CQR from Africa	—Cys—Ile —Glu—Thr—His—Ser—Glu—Ser—Ile —Ile —
CQR from Asia	—Cys—Ile —Glu—Thr—His—Ser—Glu—Ser—Thr—Ile —
CQR from Peru and Brazil	—Ser—Met—Asn—Thr—His—Ser—Gln—Asp—Leu—Arg—
CQR from Colombia	—Cys—Met—Glu—Thr—Gln—Ser—Gln—Asn—Ile —Thr—

(ii) With reference to Table 4.1, calculate the mean number of different amino acids coded for by CQR alleles in comparison with the non-CQR allele. Show your working. [2]

mean number of different amino acids

In the non-CQR *P. falciparum*, chloroquine accumulates in the digestive vacuole of *P. falciparum* and interferes with the detoxification of haem in the host red blood cell. This results in parasite death.

Research shows that there is a point mutation in the *pfcr*t gene of CQR *P. falciparum* which leads to the formation of a chloroquine-resistance transporter located in the digestive vacuole membrane of CQR *P. falciparum*.

This mutation changes the amino acid from lysine to threonine at the binding site of the chloroquine-resistance transporter.

- (iii) Using the information provided, suggest how this mutation of *pfcr*t gene increases the chloroquine resistance in *P. falciparum*. [3]

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[Total: 15]

Section B

Answer **ONE** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

You answers must be set out in parts **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 5

- (a)** Following cytokinesis, one of the daughter cells may not have a nucleolus. This cell is able to divide once more and then the new daughter cells die.

Explain how the cell is able to survive for one more cell division and suggest why the new daughter cells then die. [7]

- (b)** The same parents may produce offspring that are different in appearance. Discuss how this is brought about. [8]

[Total: 15]

QUESTION 6

- (a)** All cells, from bacteria to humans, express their genetic information via the central dogma of molecular biology which depicts the flow of genetic information from DNA to protein.

Outline the processes involved in the central dogma of molecular biology, and explain, using an example, why the central dogma may not always hold true. [7]

- (b)** The expression of genes gives rise to proteins that affect the biochemical reactions and other processes in organisms.

Discuss, with examples, the importance of specific shapes of proteins in organisms. [8]

[Total: 15]

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[illegible]

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End of Paper